

HireIQ

AI Hiring Intelligence Platform

Product Requirements Document (PRD)

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SECTION 1

Executive Summary

1. Executive Summary

HireIQ is an AI-powered SaaS hiring intelligence platform designed to transform the traditional recruitment process into a data-driven, intelligent pipeline. The platform bridges the gap between job seekers and recruiters by providing deep AI analysis of resumes, smart candidate scoring, personalized job recommendations, and automated interview evaluation.

Unlike conventional job boards, HireIQ brings ML-powered insights directly into the hiring workflow — enabling recruiters to identify the best candidates faster, and helping job seekers understand their strengths and skill gaps relative to job market demand.

1.1 Problem Statement

For Recruiters	Manually screening hundreds of resumes is time-consuming, error-prone, and subject to unconscious bias. Identifying top candidates from large applicant pools takes days.
For Job Seekers	Candidates lack visibility into why they are rejected. They have no way to measure how well their profile matches job requirements or understand which skills to improve.
For the Industry	Traditional job portals act as static notice boards. There is no intelligence layer to match candidates and jobs beyond simple keyword filtering.

1.2 Solution

HireIQ addresses these gaps by providing:

- An AI-powered Resume Intelligence Engine that extracts structured data from resumes using NLP
- A Candidate Scoring System that scores candidates 0-100 against job descriptions using semantic similarity
- A Job Recommendation System personalized to each user's profile and behavior
- An AI Interview Evaluation System that scores candidate answers and provides actionable feedback
- A Recruiter Intelligence Dashboard with analytics, applicant trends, and ranked candidate lists

- A Job Seeker Dashboard with resume score, skill gap analysis, and improvement suggestions

1.3 Product Vision

"To become the most trusted AI hiring platform that makes recruitment smarter, fairer, and faster — for both sides of the talent equation."

SECTION 2

Goals, Objectives & Success Metrics

2. Goals, Objectives & Success Metrics

2.1 Business Goals

BG-01	Achieve 10,000 active users (combined job seekers + recruiters) within 6 months of launch
BG-02	Onboard 500 active company/recruiter accounts within 6 months
BG-03	Generate \$50,000 MRR within 12 months via recruiter subscription plans
BG-04	Achieve a Net Promoter Score (NPS) of 50 or higher within 12 months
BG-05	Reduce average recruiter time-to-shortlist by 60% compared to manual screening

2.2 Product Objectives

PO-01	Build a complete full-stack platform with backend API, AI/ML services, and a modern React frontend
PO-02	Deliver AI resume scoring with 85% or higher accuracy compared to human expert evaluation
PO-03	Provide job recommendations with 70% or higher click-through relevance rate
PO-04	Ensure all pages load within 2 seconds on standard 10 Mbps broadband connections
PO-05	Maintain 99.9% API uptime SLA in production
PO-06	Pass all OWASP Top 10 security checks before launch

2.3 Key Performance Indicators (KPIs)

KPI	Metric	Target
User Acquisition	Monthly new signups	10,000 in 6 months
Engagement	Weekly active users (WAU)	40% or more of registered users
AI Accuracy	Resume scoring accuracy	85% or higher vs expert baseline
Recruiter Value	Time-to-shortlist reduction	60% faster than manual

Job Seeker Value	Application-to-interview rate	2x industry average
Platform Health	API uptime	99.9% or higher
Revenue	Monthly Recurring Revenue	\$50,000 by Month 12
Satisfaction	Net Promoter Score (NPS)	50 or above

SECTION 3

Product Scope

3. Product Scope

3.1 In Scope (v1.0)

The following modules and features are included in the initial production release:

Auth System	JWT-based registration, login, role selection (Job Seeker / Recruiter), password hashing with bcrypt
User Profiles	Profile creation and editing for both roles: skills, experience, education, profile picture upload
Job Module	Full CRUD for jobs, keyword search, filter by skills/location/salary, pagination support
Resume Module	PDF upload to AWS S3, text extraction, NLP parsing, structured data storage in PostgreSQL
Application Module	Apply to jobs, duplicate prevention, status tracking (applied/shortlisted/rejected), recruiter management
AI Resume Engine	NLP-based extraction of skills, years of experience, education level, and project names
Candidate Scoring	TF-IDF and cosine similarity scoring (0-100) with matched and missing skills breakdown
Job Recommendations	Content-based filtering using user skill profile — top 5 personalized recommendations
AI Interview	Text-based Q&A with NLP scoring, actionable feedback, and confidence labels
Recruiter Dashboard	Applicant overview, candidate ranking by AI score, hiring analytics and charts
Job Seeker Dashboard	Profile summary, AI score display, skill gap analysis, and recommended jobs feed
Admin Panel	User management, job moderation, platform analytics — v1 basic implementation

3.2 Out of Scope (v1.0) — Future Releases

Video Interviews	Live video interview recording and AI analysis — planned for v2.0
Voice Analysis	Speech-to-text tone analysis for voice interviews — v2.0
Chrome Extension	LinkedIn profile import browser extension — v2.5
Mobile Apps	Native iOS and Android applications — v2.0
ATS Integrations	Direct integrations with Workday, Greenhouse, and Lever — v2.0
Bias Audit Module	Automated bias detection and fairness reporting — v1.5
Personality Analysis	MBTI / Big Five personality scoring from text responses — v2.0
Multi-language	Non-English resume processing and platform localization — v1.5

SECTION 4

User Personas & User Stories

4. User Personas & User Stories

4.1 Persona 1 — The Job Seeker

Name	Rahul Sharma
Age	24 years old
Background	Computer Science graduate, 1 year of experience, actively job hunting
Technical Level	Intermediate — comfortable with tech but not a developer
Primary Goal	Land a software engineering role at a product company
Pain Points	Does not know why applications get rejected; unsure which skills to improve; overwhelmed by job volume
Device	Primarily mobile with occasional desktop use
Motivation	Wants AI-powered feedback to improve his profile and get noticed by recruiters

4.2 Persona 2 — The Recruiter

Name	Priya Mehta
Age	32 years old
Background	Senior HR Manager at a mid-size tech company, manages 50+ open roles per quarter
Technical Level	Low to intermediate — uses digital tools daily but is not a developer
Primary Goal	Find and shortlist top candidates faster without reading every resume manually
Pain Points	Flooded with unqualified applications; no quick ranking mechanism; lacks pipeline data
Device	Desktop primarily in an office environment
Motivation	Wants AI scoring to surface the top 10% of candidates instantly without manual review

4.3 User Stories — Job Seeker

ID	User Story	Priority
US-JS-01	As a job seeker, I want to register and select my role so the platform personalizes my experience.	Must Have
US-JS-02	As a job seeker, I want to upload my resume as a PDF so the AI can analyze my profile.	Must Have
US-JS-03	As a job seeker, I want to see my AI resume score so I know how strong my profile is.	Must Have
US-JS-04	As a job seeker, I want to see which skills I am missing for a job so I can improve.	Must Have
US-JS-05	As a job seeker, I want personalized job recommendations based on my skill profile.	Must Have
US-JS-06	As a job seeker, I want to apply for jobs and track my application status in real time.	Must Have
US-JS-07	As a job seeker, I want to take an AI mock interview to practice before the real one.	Should Have
US-JS-08	As a job seeker, I want to filter jobs by skills, location, and salary range.	Must Have
US-JS-09	As a job seeker, I want to see my profile completion percentage to stay motivated.	Should Have
US-JS-10	As a job seeker, I want toast notifications when my application status changes.	Should Have

4.4 User Stories — Recruiter

ID	User Story	Priority
US-RC-01	As a recruiter, I want to post a job with title, description, skills, and salary so candidates can apply.	Must Have
US-RC-02	As a recruiter, I want to see all applicants ranked by AI score for fast shortlisting.	Must Have
US-RC-03	As a recruiter, I want to update applicant status (shortlisted/rejected) directly from the dashboard.	Must Have
US-RC-04	As a recruiter, I want to view a candidate's parsed resume data without opening a PDF.	Must Have
US-RC-05	As a recruiter, I want applicant trend charts to understand my hiring pipeline health.	Should Have
US-RC-06	As a recruiter, I want to filter my job postings by status, date, and applicant count.	Should Have

US-RC-07	As a recruiter, I want skill distribution charts across applicants to understand the talent pool.	Should Have
US-RC-08	As a recruiter, I want to edit or delete my job postings at any time.	Must Have
US-RC-09	As a recruiter, I want to export a shortlist of top candidates to CSV.	Nice to Have

SECTION 5

Functional Requirements

5. Functional Requirements

5.1 Authentication & Authorization

FR-AUTH-01	System shall allow users to register with name, email, password, and role (job_seeker or recruiter)
FR-AUTH-02	System shall hash all passwords using bcrypt with a minimum cost factor of 12
FR-AUTH-03	System shall issue a signed JWT token on successful login with a 24-hour expiry
FR-AUTH-04	System shall implement RBAC — recruiters cannot apply to jobs, job seekers cannot post jobs
FR-AUTH-05	System shall return 401 Unauthorized for all protected routes when token is missing or invalid
FR-AUTH-06	System shall support token refresh to prevent users from being forced to re-login frequently

5.2 User Profile Module

FR-USER-01	System shall allow users to view and update their profile (name, bio, skills, experience, education)
FR-USER-02	System shall allow users to upload a profile picture in JPEG or PNG format up to 2MB
FR-USER-03	System shall calculate and display a profile completion percentage based on filled fields
FR-USER-04	System shall store skills as a structured array to enable AI matching operations

5.3 Job Module

FR-JOB-01	System shall allow authenticated recruiters to create jobs with title, description, required_skills, salary range, location, and job_type
FR-JOB-02	System shall allow any authenticated user to browse jobs with pagination (default 10 items per page)

FR-JOB-03	System shall support job search by keyword and filter by skills, location, and salary range
FR-JOB-04	System shall allow only the owning recruiter to update or delete their job posting
FR-JOB-05	System shall display creation timestamp and applicant count on each job listing card
FR-JOB-06	System shall support active/closed status and block new applications to closed jobs

5.4 Resume Module

FR-RES-01	System shall accept resume uploads in PDF format only with a maximum file size of 5MB
FR-RES-02	System shall store uploaded resumes in AWS S3 and save the public URL in the database
FR-RES-03	System shall extract text from the uploaded PDF and pass it to the AI resume engine
FR-RES-04	System shall store parsed resume data (skills, experience, education) as a structured JSON field
FR-RES-05	System shall allow users to delete their resume from both S3 storage and the database
FR-RES-06	System shall validate file type and size before any upload processing begins

5.5 Application Module

FR-APP-01	System shall allow each job seeker to apply for each job only once (duplicate application prevention)
FR-APP-02	System shall attach the user's current resume URL to the application at time of submission
FR-APP-03	System shall set application status to 'applied' by default upon submission
FR-APP-04	System shall allow the posting recruiter to update application status to 'shortlisted' or 'rejected'
FR-APP-05	System shall allow job seekers to view all their applications with current status
FR-APP-06	System shall prevent applying to a job if no resume has been uploaded by the user

5.6 AI Module

FR-AI-01	System shall expose POST /ai/analyze-resume accepting resume text, returning structured JSON with skills, experience, and education
FR-AI-02	System shall expose POST /ai/score-candidate accepting resume data and job description, returning score (0-100), matched_skills, and missing_skills
FR-AI-03	System shall expose POST /ai/recommend-jobs accepting a user_id and returning top 5 ranked job recommendations
FR-AI-04	System shall expose POST /ai/evaluate-interview accepting question and answer, returning score, feedback, and confidence level
FR-AI-05	System shall automatically trigger resume analysis and scoring on every job application submission
FR-AI-06	All AI endpoints shall respond within 5 seconds at p95 under normal production load

SECTION 6

Non-Functional Requirements

6. Non-Functional Requirements

6.1 Performance

NFR-PERF-01	All REST API endpoints (non-AI) must respond within 500ms at p95 under 1,000 concurrent users
NFR-PERF-02	AI endpoints must respond within 5 seconds at p95
NFR-PERF-03	Frontend page load time must be under 2 seconds on 10 Mbps broadband
NFR-PERF-04	Job listing API must support pagination and must not return more than 50 records per request
NFR-PERF-05	Database queries must use indexed fields for all search and filter operations

6.2 Scalability

NFR-SCALE-01	Backend must be stateless and horizontally scalable using Docker containers
NFR-SCALE-02	ML service must run as a separate deployable microservice for independent scaling
NFR-SCALE-03	Redis caching must be used for job listings, user profiles, and AI scores to reduce DB load
NFR-SCALE-04	S3 storage must handle resume uploads without any server-side file processing bottlenecks

6.3 Security

NFR-SEC-01	All passwords must be hashed using bcrypt — never stored in plain text under any circumstance
NFR-SEC-02	All API routes must be JWT-protected except public job listings and auth endpoints
NFR-SEC-03	Application must pass all OWASP Top 10 checks before production release
NFR-SEC-04	All user inputs must be sanitized to prevent SQL injection and XSS attacks

NFR-SEC-05	S3 bucket must not be publicly writable — all uploads must flow through pre-signed URLs
NFR-SEC-06	Rate limiting must be enforced at 100 requests/min per IP on authentication endpoints

6.4 Reliability & Availability

NFR-REL-01	System must target 99.9% uptime — fewer than 8.7 hours of unplanned downtime per year
NFR-REL-02	Database must have automated daily backups with 30-day retention policy
NFR-REL-03	All API errors must be logged with request ID, timestamp, and stack trace
NFR-REL-04	Health check endpoints must be exposed on all services for load balancer integration

6.5 Usability

NFR-UX-01	All UI pages must be fully responsive and functional on screen widths from 320px to 2560px
NFR-UX-02	All interactive elements must provide visual feedback within 200ms of user action
NFR-UX-03	All form validation errors must be displayed inline in real time — not only on form submit
NFR-UX-04	Empty states must be shown for all list views when no data is available
NFR-UX-05	All async actions must display loading indicators such as spinners or skeleton loaders

SECTION 7

System Modules & Feature Breakdown

7. System Modules & Feature Breakdown

The platform is composed of 8 core modules. Each has a defined scope and priority classification.

Module	Description	Priority
M01 — Auth	JWT registration/login, RBAC, password hashing, token refresh	P0 — Critical
M02 — User	Profile management, skills/experience, avatar upload, completion score	P0 — Critical
M03 — Jobs	CRUD for jobs, search, filter, pagination, job status management	P0 — Critical
M04 — Resume	PDF upload to S3, text extraction, structured data parsing and storage	P0 — Critical
M05 — Applications	Apply to jobs, status tracking, recruiter updates, duplicate prevention	P0 — Critical
M06 — AI Engine	Resume NLP analysis, candidate scoring, recommendations, interview eval	P1 — High
M07 — Dashboards	Role-based dashboards, analytics charts, candidate rankings, insights	P1 — High
M08 — Admin	User/job moderation, platform analytics, flagging and reporting	P2 — Medium

7.1 AI Module Deep Dive

The AI Engine (M06) is the core differentiator of this platform. It consists of four integrated sub-systems:

Resume Intelligence Engine	Accepts PDF text. Uses spaCy NER and custom skill dictionaries to extract: skills array, years_of_experience, education_level, and projects list. Returns structured JSON stored in the resumes table.
Candidate Scoring System	Accepts resume JSON and job description. Vectorizes both using TF-IDF. Computes cosine similarity (0.0-1.0) mapped to a 0-100 score. Also performs skill gap analysis: matched_skills and missing_skills arrays returned alongside the score.

Job Recommendation System	Content-based filtering: vectorizes the user skill profile and computes similarity against all active job descriptions. Returns top 5 job IDs ranked by similarity score. Refreshed on every profile update.
AI Interview Evaluator	Accepts interview question and candidate text answer. Compares against a curated ideal answer corpus. Returns score (0-100), detailed feedback string, and a confidence label: low, medium, or high.

SECTION 8

Information Architecture & Data Model

8. Information Architecture & Data Model

8.1 Database Tables Overview

Table	Key Fields	Notes
users	id, name, email, password_hash, role, bio, skills[], experience, education, avatar_url, profile_complete_pct, created_at, updated_at	Core user entity for both roles
jobs	id, title, description, required_skills[], salary_min, salary_max, location, job_type, status, recruiter_id (FK), applicant_count, created_at, updated_at	One recruiter maps to many jobs
resumes	id, user_id (FK unique), file_url, raw_text, parsed_skills[], parsed_experience, parsed_education, ai_score, created_at, updated_at	One user has one active resume
applications	id, user_id (FK), job_id (FK), resume_url, ai_score, status, applied_at, updated_at	Many-to-many resolved via join table
ai_results	id, user_id (FK), job_id (FK), score, matched_skills[], missing_skills[], recommendation_data, created_at	AI scoring and recommendation cache
interview_sessions	id, user_id (FK), job_id (FK nullable), questions[], answers[], scores[], overall_score, completed_at	AI mock interview history

8.2 Entity Relationships

users to jobs	One-to-many: One recruiter can post many jobs (recruiter_id FK on jobs table)
users to resumes	One-to-one: Each job seeker has one active resume (user_id UNIQUE on resumes table)
users to applications	One-to-many: One user can apply to many jobs (user_id FK on applications)
jobs to applications	One-to-many: One job can receive many applications (job_id FK on applications)
users and jobs	Many-to-many relationship resolved through the applications join table

users to sessions	One-to-many: One user can have many AI interview practice sessions
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8.3 Application Route Architecture

Route	Description	Access
/	Landing page	Public
/auth/login	Login page	Public
/auth/register	Registration and role selection	Public
/dashboard	Role-based dashboard (redirects by role)	Authenticated
/jobs	Job listing with search and filter	Authenticated
/jobs/:id	Full job detail view	Authenticated
/jobs/create	Create new job posting form	Recruiter only
/resume	Resume upload and management	Job Seeker only
/applications	My applications with status tracker	Job Seeker only
/ai/insights	AI score and skill gap analysis	Job Seeker only
/ai/interview	AI mock interview interface	Job Seeker only
/recruiter/applicants/:jobId	All applicants for a specific job	Recruiter only
/recruiter/analytics	Hiring analytics dashboard with charts	Recruiter only
/profile	Edit profile page	Authenticated
/admin	Admin moderation panel	Admin only

SECTION 9

Acceptance Criteria

9. Acceptance Criteria

The following criteria must be met before each module is considered production-ready:

9.1 Authentication

- User can register with valid data and receive a JWT token in the response
- Duplicate email registration returns HTTP 409 Conflict with a clear error message
- Invalid login credentials return HTTP 401 Unauthorized
- Protected endpoints return 401 when called without a valid token
- Role-based routes return HTTP 403 Forbidden when accessed by the wrong role

9.2 Job Module

- Recruiter can create a job and it appears in the public job listing immediately
- Job search by keyword returns relevant results within 1 second
- Pagination works correctly — page 1 and page 2 return non-overlapping results
- Only the owning recruiter can edit or delete their job — others receive HTTP 403
- Applying to a closed job returns HTTP 400 with an appropriate error message

9.3 Resume & AI

- Uploading a non-PDF file returns a validation error with a file type message
- Uploading a file larger than 5MB returns a file size error before processing begins
- After successful upload, parsed skills appear in the user profile within 10 seconds
- AI score is generated and stored automatically for every new job application
- Score returned by `/ai/score-candidate` is always an integer between 0 and 100
- `Missing_skills` array correctly identifies skills in the job description absent from the resume

9.4 Dashboard & UI

- Recruiter dashboard shows all applicants ranked by AI score in descending order

- Charts render correctly on both desktop (1280px+) and mobile (375px) viewports
- All loading states display skeleton loaders or spinners until data is fully fetched
- Dark mode toggle persists correctly across page navigation and browser refresh
- All toast notifications auto-dismiss after 4 seconds without user interaction

SECTION 10

Release Roadmap

10. Release Roadmap

10.1 Development Phases

Phase	Timeline	Deliverables
Phase 1 — Foundation	Weeks 1-3	Project setup, DB schema, Auth module, User module, CI/CD pipeline
Phase 2 — Core Features	Weeks 4-6	Jobs module, Resume upload with S3, Application module, basic dashboards
Phase 3 — AI Integration	Weeks 7-9	NLP resume engine, candidate scoring, job recommendations, all AI API endpoints
Phase 4 — Frontend Polish	Weeks 10-11	AI Insights page, AI Interview UI, Recharts analytics, dark mode, full responsiveness
Phase 5 — QA & Security	Week 12	End-to-end testing, OWASP Top 10 checks, performance load testing, bug fixes
Phase 6 — Launch	Week 13	Production deployment to AWS, monitoring, documentation, soft launch to beta users

10.2 Post-Launch Iterations

v1.1 — Month 3	Bias reduction module, resume version history, email notifications, recruiter CSV export
v1.5 — Month 6	Multi-language resume support, advanced filters, calendar integration for interview scheduling
v2.0 — Month 12	Native mobile apps (iOS + Android), video interview recording, AI personality analysis, ATS integrations

SECTION 11

Risks, Assumptions & Dependencies

11. Risks, Assumptions & Dependencies

11.1 Risk Register

Risk	Severity	Mitigation Strategy
AI scoring accuracy too low	High	Benchmark with expert-labeled data early; set minimum accuracy threshold before launch
S3 upload latency under load	Medium	Use pre-signed URLs and async upload with real-time progress tracking in UI
JWT token theft or auth issues	High	HTTPS only, short token expiry, refresh token rotation, secure httpOnly cookies
ML inference too slow at scale	Medium	Cache AI scores in Redis; run inference asynchronously after application submission
GDPR / data privacy gaps	High	Implement data deletion API, privacy policy, and consent flags before launch
Scope creep delaying v1	Medium	Strict feature freeze after Phase 2; extra requests go into the v2 roadmap

11.2 Assumptions

- Users have access to modern browsers — Chrome, Firefox, and Safari within the last 2 major versions
- AWS credentials and S3 bucket will be provisioned and available by Week 2 of development
- The ML team will integrate real models by Phase 3 — Phase 2 uses documented mock responses
- PostgreSQL will be hosted on a managed service such as AWS RDS or Supabase
- All team members have access to a shared development environment via Docker Compose

11.3 External Dependencies

AWS S3

Resume file storage — required for all production environments

AWS RDS or Supabase	Managed PostgreSQL database — required for production
Redis Cloud or ElastiCache	Distributed caching — required to meet performance targets
spaCy or Hugging Face	NLP models powering the resume intelligence engine
scikit-learn	TF-IDF vectorizer and cosine similarity — required for candidate scoring
Recharts or Chart.js	Frontend data visualization — required for all dashboard analytics

SECTION 12

Glossary & Appendix

12. Glossary

PRD	Product Requirements Document — the master specification for a software product
JWT	JSON Web Token — a compact token used for stateless authentication
RBAC	Role-Based Access Control — restricting access based on user roles
NLP	Natural Language Processing — AI technique for understanding human text
NER	Named Entity Recognition — NLP sub-technique to extract named entities from text
TF-IDF	Term Frequency-Inverse Document Frequency — statistical text vectorization method
Cosine Similarity	Measure of similarity between two vectors — used for resume-to-job matching
S3	Amazon Simple Storage Service — cloud object storage used for resume files
MRR	Monthly Recurring Revenue — the primary revenue KPI for SaaS businesses
NPS	Net Promoter Score — customer satisfaction metric ranging from -100 to +100
p95	95th percentile — the latency within which 95% of all requests complete
OWASP	Open Web Application Security Project — industry standard web security guidelines
ATS	Applicant Tracking System — software recruiters use to manage job applications
WAU	Weekly Active Users — users who engage at least once per week
KPI	Key Performance Indicator — a measurable value showing how effectively goals are being met

HireIQ PRD v1.0 — End of Document

This document is the single source of truth for HireIQ v1.0. All subsequent technical documents derive from this PRD.